**3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity**

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

Answer:(penalty regime: 0 %)

**Feedback**

```
#include <stdio.h>


int main()
```

```
{
   int T;
   scanf("%d", &T);

   while (T--)
   {
      int m, n;

      scanf("%d", &m);

      int a[m];

      for (int i = 0; i < m; i++)
         scanf("%d", &a[i]);

      scanf("%d", &n);

      int b[n];

      for (int i = 0; i < n; i++)
         scanf("%d", &b[i]);

      int first = 1;

      for (int i = 0; i < m; i++)
      {
         for (int j = 0; j < n; j++)
         {
```

```
            if (a[i] == b[j])
            {
               if (!first)
                  printf(" ");

               printf("%d", a[i]);
               first = 0;
               break;
            }
         }
      }

      printf("\n");
   }

   return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | |
| | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.